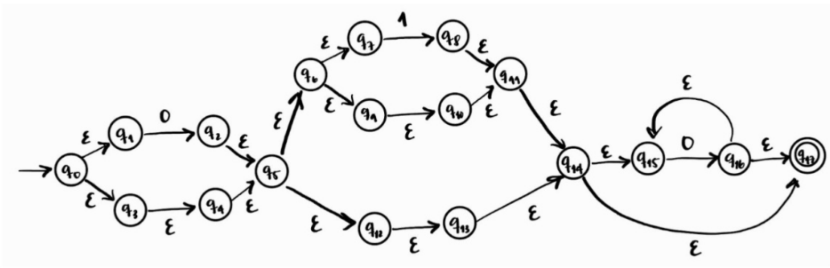
Problema 2: 25%

Utilice el Lema de Arden para encontrar la expresión regular del siguiente autómata. Deje todo su procedimiento



$A = aA \mid b = a^*b$

$q_0 = \varepsilon q_1 \mid \varepsilon q_3$ 1

$q_1 = 0q_2$ 2

$q_2 = \varepsilon q_5 = q_4$ 3 y 4

$q_3 = \varepsilon q_4$ 5

$q_5 = \varepsilon q_6 \mid \varepsilon q_{12}$ 6

$q_6 = \varepsilon q_7 \mid \varepsilon q_9$ 7

$q_7 = 1q_8$ 8

$q_8 = \varepsilon q_{11} = q_{10}$ 9 y 10

$q_9 = \varepsilon q_{10}$ 11

$q_{11} = \varepsilon q_{14}$ 12

$q_{12} = \varepsilon q_{13}$ 13

$q_{13} = \varepsilon q_{14}$ 14

$q_{14} = \varepsilon q_{15} \mid \varepsilon q_{17}$ 15

$q_{15} = 0q_{16}$ 16

$q_{16} = \varepsilon q_{15} \mid \varepsilon q_{17}$ 17

$q_{17} = \varepsilon$ 18

$\Rightarrow q_{16} = \varepsilon q_{15} \mid \varepsilon\varepsilon = \varepsilon q_{15} \mid \varepsilon$

$q_{16} = \varepsilon 0 q_{16} \mid \varepsilon = 0q_{16} \mid \varepsilon = 0^*\varepsilon = 0^*$ (Lema Arden)

$\Rightarrow q_{15} = 00^* = 0^+$

$\Rightarrow q_{14} = (\varepsilon 0^+) \mid \varepsilon\varepsilon = 0^+ \mid \varepsilon = 0^*$

$\Rightarrow q_{13} = \varepsilon(0^*) = 0^*$

$\Rightarrow q_{12} = \varepsilon(0^*) = 0^*$

$\Rightarrow q_{11} = \varepsilon(0^*) = 0^*$

$\Rightarrow q_{10} = \varepsilon(0^*) = 0^* = q_8$

$\rightarrow q_9 = \varepsilon(0^*) = 0^*$

$\Rightarrow q_7 = 10^*$

$\Rightarrow q_6 = \varepsilon(10^*) \mid \varepsilon 0^* = (10^*) \mid 0^*$

$\Rightarrow q_5 = \varepsilon[(10^*) \mid 0^*] \mid \varepsilon 0^* = [(10^*) \mid 0^*] \mid 0^*$

$\Rightarrow q_4 = \varepsilon\{[(10^*) \mid 0^*] \mid 0^*\} = [(10^*) \mid 0^*] \mid 0^* = q_2$

$\Rightarrow q_3 = \varepsilon\{[(10^*) \mid 0^*] \mid 0^*\} = [(10^*) \mid 0^*] \mid 0^*$

$\Rightarrow q_1 = 0\{[(10^*) \mid 0^*] \mid 0^*\} = 0[(10^*) \mid 0^*] \mid 00^* = [(010^*) \mid 0^+] \mid 0^+$

$\rightarrow q_0 = \varepsilon\{[(010^*) \mid 0^+] \mid 0^+\} \mid \varepsilon\{[(10^*) \mid 0^*] \mid 0^*\} = [[(010^*) \mid 0^+] \mid 0^+] \mid [(10^*) \mid 0^*]$